

IDC 410: Machine Learning

Spring Semester 2025-26

Assignment 10

Instructor: Prof. Shashikant R. Dugad

Assignment prepared by: Siddharth

Start Date: 12th April 2026 (Sunday)

Submission: Before 19th April 2026 (Sunday) 23:00 hrs

1. Install **VOC2012** dataset on your laptop (You may copy the dataset from me or Siddharth on your pendrive). Use **YOLOv8** to train and identify objects in this dataset. Describe the results of your evaluation.
2. Load the **MUTAG** dataset using PyTorch Geometric. Visualize the molecular graphs for at least 4 sample molecules (both mutagenic and non-mutagenic examples). Design and train a Graph Neural Network (GNN) of your choice (e.g. GCN, GAT, GraphSAGE, GIN) to classify molecules in the MUTAG dataset as mutagenic or non-mutagenic.
3. (a) For each molecule in the MUTAG dataset, compute the number of 6-membered rings as a new annotation. You may use an automated/manual approach.

(b) Using these ring counts as target labels, train a GNN to predict the number of 6-membered rings in a molecule. You may treat this as a regression or a multi-class classification problem; justify your choice. Report appropriate evaluation metrics on a held-out test set.

Please NOTE:

1. Complete your assignment in time
2. Contact Siddharth for any clarification.
3. If you need any clarifications, feel free to contact TA's or me well in advance so that you can complete your assignment in time